Week 9: 26-30 October

Subject:Computer science | Unit Title عنوان الوحدة: Computer fundamentals | Grade الصف: 11DP SL

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
الحصة **Lesson 1: Logic gates, truth tables and simple diagrams Syllabus references: A1.2.3, A1.2.4, A1.2.5 (Part 1)**	• I can identify and describe the output rules for AND, OR, NOT, NAND, NOR, XOR and XNOR gates. • I can construct complete truth tables for single gates and simple two-input Boolean expressions. • I can translate a short system requirement into a Boolean expression and a simple logic diagram.	**Starter / engagement — 0–6 min** Human-gate challenge: students use 0/1 cards to predict outputs for AND, OR and NOT; teacher reveals the rule. **Teacher instruction — 6–18 min** Model 0=False and 1=True; introduce the seven gate rules, standard symbols and key wording: both/all, at least one, not, different and same. **Guided practice — 18–30 min** Complete and check single-gate truth tables; contrast OR with XOR and identify inversion bubbles for NAND, NOR and XNOR. **Student-centred task — 30–48 min** Gate-station rotation: match scenarios, Boolean expressions, symbols and truth tables; then draw one simple circuit from a requirement. **Formative assessment — 48–55 min** Mini-whiteboard hinge questions and peer checking against gate rules. **Plenary — 55–60 min** Exit ticket: select a gate, complete one truth-table row and justify the choice in one sentence. **Differentiation** Support: gate-rule mat, partially completed tables and paired verbal rehearsal. Extension: explain why NAND and NOR are universal gates and build NOT using one universal gate.	**Resources** • IB Computer Science guide and A1.2.3–A1.2.5 teaching resources. • Gate-symbol cards, 0/1 cards, mini-whiteboards and truth-table worksheet. • Logic simulator, if available, for checking simple circuits. • Parent link: student lesson notes and completed gate summary table.
Lesson 2: Combined circuits, equivalence and simplification	• I can analyse a multi-gate logic diagram using intermediate outputs and construct its complete truth table. • I can convert accurately between a problem description, Boolean expression, logic diagram and truth table.	**starter — 0–6 min** Gate grid: identify seven gates and correct one OR/XOR misconception. **Teacher instruction — 6–19 min**	**. Resources** • IB Computer Science guide and A1.2.3–A1.2.5 teaching resources. • Boolean-law reference sheet, K-map grids and circuit-design brief.

Syllabus references: A1.2.3, A1.2.4, A1.2.5 (Part 2)	• I can simplify a Boolean expression using appropriate laws or a K-map and verify logical equivalence.	Model Q=(A AND B) OR C from left to right. Use one intermediate column per gate and 2^n to determine the number of truth-table rows. **Deepening instruction — 19–29 min** Introduce identity, null, idempotent, complement, double negation, distributive and De Morgan's laws; demonstrate a two-variable K-map and grouping rules. **Student-centred application — 29–48 min** Circuit design challenge: from a safety-system brief, define inputs, write the expression, draw the circuit, create the truth table, simplify where possible and redraw. **Formative assessment — 48–55 min** Gallery walk: peers verify gate order, intermediate columns, all input combinations and matching outputs. **Plenary — 55–60 min** IB-style exit question: explain how a truth table proves that two circuits are logically equivalent. **Differentiation** Support: colour-coded stages, pre-labelled inputs and partially completed 3-input tables. Extension: compare algebraic and K-map simplification, then justify which representation communicates the solution most clearly.	• Logic simulator for testing multi-gate circuits and comparing outputs. • Parent link: revision sheet covering truth tables, Boolean notation and simplification laws.
Lesson 3: Role of operating systems **Syllabus references: A1.3.1**	• I can describe how an operating system provides an interface between users, applications and hardware. • I can explain hardware abstraction using a clear example such as saving a file or printing. • I can analyse how an operating system manages CPU time, RAM, storage and input/output resources in a multitasking scenario.	**. Starter / engagement — 0–6 min** What happens when you click Save? Students sequence user, application, OS, storage management and storage device. **Teacher instruction — 6–20 min** Model the OS as resource manager and hardware-abstraction layer. Explain CPU scheduling, memory allocation, file/storage services and input/output management. **Student-centred jigsaw — 20–39 min** Expert groups investigate CPU, RAM, storage or I/O management, then teach mixed groups using one precise example and one consequence if management failed. **Application task — 39–50 min** Multitasking case study: browser, music player, word processor and background services compete for resources; groups map OS decisions and interactions. **Formative assessment — 50–55 min** Concept-check questions distinguish resource management from hardware abstraction. **Plenary — 55–60 min**	• IB Computer Science guide and A1.3.1 teaching resource. • OS layer diagram, resource-management cards and multitasking case study. • Device examples: printer, storage drive, keyboard, monitor and network interface. • Parent link: A1.3.1 vocabulary and resource-management summary.

		Exit ticket: explain in three linked points how an OS allows an application to save a file without controlling the hardware directly. **Differentiation** Support: labelled layer diagram, key-term bank and sentence stems. Extension: evaluate how a standardised OS interface reduces the need to rewrite applications for different hardware.	
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Prior knowledge and progression** Students should recall binary states, Boolean values and basic data representation. Lesson 1 establishes gate behaviour and simple representations; Lesson 2 integrates and simplifies combined logic; Lesson 3 applies abstraction and resource-management thinking to system software. **Checks before teaching** Confirm students use 2^n rows, distinguish OR from XOR, read diagrams left to right, label intermediate outputs and treat Boolean + and · as logical operators. **Evidence to retain** Exit tickets, completed truth tables, circuit-design artefact and A1.3.1 case-study response.		

Week 9

Technology Integration دمج التكنولوجيا	Use an interactive logic-circuit simulator in Lessons 1–2 to construct and verify circuits; use collaborative digital annotation in Lesson 3 to map OS resource allocation.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: powers of two, symbolic notation and equivalence. Physics/electronics: binary signals and circuits. Design technology: modelling systems against requirements.
External Assessment Integration دمج التقييمات الخارجية	Use IB-style command terms identify, construct, describe, explain, analyse and evaluate. Require complete truth tables, visible intermediate working and concise cause-and-effect explanations.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: analysis and transfer between representations. Communication: precise technical vocabulary. Social: structured pair and jigsaw roles. Self-management: timed tasks and checking routines.
Standards المعايير	A1.2.3 logic gates and Boolean operators; A1.2.4 truth tables and logic simplification; A1.2.5 construction and analysis of logic diagrams; A1.3.1 role of operating systems.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	How does the representation chosen — words, Boolean expressions, truth tables or diagrams — shape what is easy to understand or verify? What knowledge is hidden when an OS abstracts hardware?
Success Criteria معايير النجاح (Measurable)	Lesson 1 — MUST: Correctly identify all seven gate rules and complete a simple truth table. SHOULD: Translate a requirement into a matching expression and diagram. COULD: Construct NOT using NAND or NOR and justify the result. Lesson 2 — MUST: Complete a combined-circuit truth table using intermediate columns. SHOULD: Convert accurately among a brief, expression, diagram and truth table. COULD: Simplify a circuit and prove equivalence using matching outputs. Lesson 3 — MUST: Identify the OS as the layer managing hardware and software resources. SHOULD: Explain CPU, memory, storage and I/O management with examples. COULD: Analyse how abstraction supports the same application across different hardware.
National Identity Domain مجالات الربط بالهوية الوطنية	Innovation and future readiness: connect digital logic and operating-system design to reliable, efficient computing systems used in the UAE.
	Responsibility and cybersecurity: discuss how accurate logical conditions support safe access-control and monitoring systems.

	Community and collaboration: use peer explanation, respectful critique and shared problem solving during circuit and OS case-study tasks.